

Sicilian–English Neural Machine Translation: an Automated Data Pipeline and Reproducible Baselines

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Abstract

Sicilian is a low-resource Romance language, with a literary tradition older than Italian, yet absent from every major machine-translation service. Building on Wdowiak’s *Tradutturi Sicilianu* [Wdowiak, 2021], the first Sicilian neural machine translation (NMT) system, we rebuild the parallel-corpus pipeline with sentence-embedding extraction and alignment (PyMuPDF and LaBSE [Feng et al., 2022]), assemble a reproducible, deduplicated Sicilian–English dataset with a held-out *literary* test set, and port the paper’s tokenization and desinence-biasing recipe to a Perl-free stack. We evaluate both from-scratch Sockeye baselines and the pretrained NLLB-200 model [NLLB Team, 2022] adapted with LoRA [Hu et al., 2021]. On our harder held-out test, Sicilian→English climbs from a 5.27 BLEU floor to 31.33 with a bidirectionally fine-tuned NLLB-1.3B—above Wdowiak’s published baseline (29.1)—while the reverse direction nearly doubles (9.89→18.65). The full pipeline, dataset assembly and evaluation are reproducible.

1 Introduction

Sicilian is a Romance language spoken daily by millions in Sicily, Calabria and Puglia, and in diaspora communities abroad. The Sicilian School of poets at the court of Frederick II created the first literary standard in Italy in the 13th century and inspired Dante; Sicilian thus emerged as a literary language *before* Italian. Despite this, it is absent from Google Translate, Bing, Yandex and DeepL—as are most regional languages.

The bottleneck was never the model but the *parallel text*: nobody had assembled it. The material exists, in the bilingual literary journal published for over forty years by Arba Sicula. Wdowiak’s *Tradutturi Sicilianu* [Wdowiak, 2021] was the first Sicilian NMT system, trained on hand-aligned Arba Sicula text with a recipe tailored to the low-resource, morphologically rich setting. This work rebuilds that effort with maintained, mostly CPU-only tooling and pushes quality with modern pretrained models.

Contributions.

- An automated extraction and alignment pipeline for the *Arba Sicula* PDFs (PyMuPDF and LaBSE [Feng et al., 2022]), replacing the legacy pdflatex+hunalign flow, with thresholds tuned against Wdowiak’s hand-aligned gold.
- A reproducible, deduplicated, provenance-tracked dataset with a held-out literary test set (not FLORES), and a BLEU/chrF evaluation harness.
- The paper’s “lever B” recipe (Napizia tokenizer + desinence-biased subwords) ported to a Perl-free pipeline, and a linguistic-feature lever (lemma source factors).
- NLLB-200 zero-shot and LoRA fine-tuning that beats the published baseline on our harder test set, with a bidirectional adapter that also rescues the reverse direction.

2 Background

NMT models translation as a conditional language model: an encoder reads the source $x = (x_1, \dots, x_n)$ and a decoder factorises the target left to right, $p_\theta(y \mid x) = \prod_t p_\theta(y_t \mid y_{<t}, x)$, trained by maximising the log-likelihood of a parallel corpus and decoded with beam search. The dominant parameterisation is the **Transformer** [Vaswani et al., 2017], which replaces recurrence with scaled dot-product attention, $\text{Attention}(Q, K, V) = \text{softmax}(QK^\top / \sqrt{d_k})V$, run in parallel heads.

With little data the default Transformer overfits; the low-resource recipe of Sennrich and Zhang [2019] prescribes a smaller, heavily regularised model with high dropout [Srivastava et al., 2014] and small subword vocabularies. **Subword segmentation** via byte-pair encoding [Sennrich et al., 2016] gives an open vocabulary and shares statistics across word forms—essential for a morphologically rich language. Further gains come from **linguistic input features** [Sennrich and Haddow, 2016], **back-translation** of monolingual data [Sennrich et al., 2015], and **multilingual** training with a target-language token [Johnson et al., 2016] that enables bridging through related high-resource languages [Fan et al., 2020, Arivazhagan et al., 2019]. Low-resource MT remains hard for exactly these reasons [Koehn and Knowles, 2017].

3 Data Pipeline

3.1 Extraction from PDFs

We extract clean text from the Arba Sicula issue PDFs with PyMuPDF (no pdflatex round-trip). Each page is assigned a language from its share of Sicilian stop-words and diacritics; the journal sets Sicilian and English on facing pages, giving a natural even/odd prior, so we form facing candidate pairs. Pages are segmented into sentences with `pysbd` (per language), filtering “furniture” (all-caps headers, page numbers, captions). The procedure recovers parallel pages the manual selection had skipped.

3.2 Sentence alignment

We embed every sentence with LaBSE [Feng et al., 2022], a language-agnostic dual encoder, and build the cosine-similarity matrix $S_{ij} = \langle u_i, v_j \rangle / (\|u_i\| \|v_j\|)$. A facing pair is confirmed only if its mean best-match similarity exceeds τ_{page} ; sentences are then aligned by a monotonic dynamic program over S allowing 1–1, 1–2 and 2–1 moves with a null penalty, in the spirit of Vecalign [Thompson and Koehn, 2019] but without a bilingual dictionary or hunalign. The thresholds $(\tau_{\text{page}}, \tau_{\text{sent}}) = (0.50, 0.40)$ were tuned against Wdowiak’s hand-aligned gold (issues AS41–42); the pipeline recovers 14,245 sentence pairs from 44 issues, comparable to his manual alignment. Crucially, the *unaligned* English is retained as an in-domain monolingual pool for back-translation.

3.3 Additional sources

We add the author’s curated “Good-Sicilian-in-NLLB” set—LASER3-mined sentences [Heffernan et al., 2022, NLLB Team, 2022] rescored and filtered for quality and Corsican contamination—a small curated “Good-Sicilian-from-WikiMatrix” set, and the WikiMatrix it-scn subset [Schwenk et al., 2021] as an auxiliary Italian bridge.

3.4 Unified dataset

Sources are cleaned, deduplicated across one another, and split deterministically. A 1,000/1,000 valid/test split is *frozen* once, held out from Arba Sicula (a literary standard, not FLORES), so every later result stays comparable (Table 1).

4 Tokenization and Subwords

We re-implement the Napizia tokenizer in pure Python (verified identical to the original Perl): it ASCII-folds accents and *uncontracts* Sicilian (e.g.

Table 1: Unified dataset.

split	pairs	source
train	27,386	Arba Sicula + NLLB
valid	1,000	Arba Sicula
test	1,000	Arba Sicula (literary)
it-scn (aux)	11,389	WikiMatrix

$\hat{o} \rightarrow a\ lu$, $\hat{d} \rightarrow di\ lu$), cutting sparsity. “Lever B” then biases the joint scn-en BPE [Sennrich et al., 2016] toward the textbook *desinenches* catalogued in the resources Wdowiak assembled to standardise Sicilian—Dieli’s *Sicilian Vocabulary*, Cipolla’s *Mparamu lu sicilianu* [Cipolla, 2013], Bonner’s grammar [Bonner, 2001], and the *Chiù dâ Palora* dictionary—so morphological endings survive as units. BPE-dropout [Provilkov et al., 2020] adds subword regularisation by sampling segmentations during training.

5 Models

The from-scratch baseline is a Sockeye-3 [Hieber et al., 2017] small Transformer (3 layers, model size 256, 4 heads, feed-forward 1024, weight tying, dropout 0.4, joint BPE $\approx 4k$; $\sim 6.6M$ parameters), trained on CPU. On top of it we stack lever B (tokenization + desinenches), more parallel text, and “lever D” linguistic input features [Sennrich and Haddow, 2016]: a lemma source factor summed into the source embedding, $e_j = E_{\text{word}}(w_j) + E_{\text{lemma}}(\ell_j)$, whose small (~ 900) vocabulary generalises across inflected forms.

The pretrained model is NLLB-200 [NLLB Team, 2022], a massively multilingual encoder-decoder that supports Sicilian (`scn_Latn`) and is steered by forcing the first decoder token. We adapt the 1.3B variant with LoRA [Hu et al., 2021]: freezing W_0 and learning a low-rank update $W = W_0 + \frac{\alpha}{r}BA$ ($r=32$, $\alpha=64$) on the attention projections, with the base in fp16 and the adapters in fp32. Training *both* directions at once (scn $\rightarrow$ en and en $\rightarrow$ scn) yields a single adapter that serves both and rescues the weak reverse direction.

6 Evaluation

We score with sacreBLEU—BLEU (`tok:13a`) and chrF—on the frozen literary test set rather than the NLLB/FLORES set: the FLORES Sicilian uses the post-2017 orthography and a different (non-literary) domain, so it is not representative of the Arba Sicula material. The Sockeye rows are scored in tokenized space (the same space as the upstream system; the raw-text floor is 5.27 BLEU), while the NLLB rows are scored on raw text.

Table 2: Our models, our test set (scn→en).

model	BLEU	chrF
floor (copy source)	5.27	25.40
Sockeye baseline	5.54	28.28
+ lever B (tok. + desinences)	7.24	29.52
+ more data	9.79	33.82
+ lever D (lemma source factors)	10.85	35.22
NLLB-200 1.3B, zero-shot (raw)	29.00	55.23
NLLB-200 1.3B, LoRA fine-tuned, scn→en only	31.16	56.79
NLLB-200 1.3B, LoRA <i>bidirectional</i>	31.33	56.92

7 Results

On the frozen held-out test set (1,000 literary pairs), Sicilian→English results are reported in Table 2. The Sockeye levers stack from 5.54 to 10.85 BLEU on CPU alone; the pretrained NLLB-1.3B wins decisively even zero-shot (29.00), and LoRA fine-tuning lifts it to 31.33—above Wdowiak’s published baseline (29.1) on our harder test.

In the reverse direction (en→scn), the same bidirectional adapter lifts NLLB-1.3B from 9.89 (zero-shot) to 18.65 BLEU / 49.82 chrF, nearly doubling the weak direction at no cost to scn→en and giving a usable two-way model. This is still below the published en→scn baseline (25.1); back-translation and Italian bridging are the next levers.

Relation to prior published numbers. Wdowiak reports, on his own in-domain hand-selected test set with the full recipe, En→Sc/Sc→En BLEU of 25.1/29.1 (baseline), 35.0/36.8 (augmented), and 45.1/48.6 (Reverse-Training). These are *not* a head-to-head with ours: the test sets and pipelines differ. A fair comparison requires running one system on the other’s test set, or reproducing the full recipe on our data—the goal of the next experiments. That our fine-tuned NLLB already clears his *baseline* on a harder, out-of-domain literary test is nonetheless a meaningful signal.

8 Conclusion and Future Work

We presented a reproducible, Perl-free path from Arba Sicula PDFs to a Sicilian–English NMT system: an automated extraction-and-alignment pipeline, the paper’s tokenization and desinence recipe with an added linguistic-feature lever, and NLLB-200 adapted with LoRA. On our harder literary test set the best model reaches 31.33 BLEU (scn→en), above the published baseline, with a usable reverse direction. Ongoing work adds **back-translation** from

the in-domain monolingual English harvested from unaligned text, **Italian bridging** through the it-scn data [Fan et al., 2020], and **serving** the model as a local web application and a Telegram bot (a FastAPI service over the NLLB adapter). All code, the data pipeline and the evaluation harness are released for reproducibility.

References

- Naveen Arivazhagan, Ankur Bapna, Orhan Firat, Dmitry Lepikhin, Melvin Johnson, Maxim Krikun, Mia Xu Chen, Yuan Cao, George F. Foster, Colin Cherry, Wolfgang Macherey, Zhifeng Chen, and Yonghui Wu. “Massively Multilingual Neural Machine Translation in the Wild: Findings and Challenges”. *arXiv*, 2019. URL <http://arxiv.org/abs/1907.05019>.
- J. K. Bonner. *Introduction to Sicilian Grammar*. Legas, Brooklyn, NY, 2001.
- Gaetano Cipolla. *Learn Sicilian, Mparamu lu sicilianu*. Legas, Mineola, NY, 2013.
- Angela Fan, Shruti Bhosale, Holger Schwenk, Zhiyi Ma, Ahmed El-Kishky, Siddharth Goyal, Mandeep Baines, Onur Celebi, Guillaume Wenzek, Vishrav Chaudhary, Naman Goyal, Tom Birch, Vitaliy Liptchinsky, Sergey Edunov, Edouard Grave, Michael Auli, and Joulin Armand. “Beyond English-Centric Multilingual Machine translation”. *arXiv*, 2020. URL <https://arxiv.org/abs/2010.11125>.
- Fangxiaoyu Feng, Yinfei Yang, Daniel Cer, Naveen Arivazhagan, and Wei Wang. Language-agnostic BERT sentence embedding. In *ACL*, 2022.
- Kevin Heffernan, Onur Çelebi, and Holger Schwenk. Bitext mining using distilled sentence representations for low-resource languages. *arXiv preprint arXiv:2205.12654*, 2022.
- Felix Hieber, Tobias Domhan, Michael Denkowski, David Vilar, Artem Sokolov, Ann Clifton, and Matt Post. “Sockeye: A Toolkit for Neural Machine Translation”. *arXiv*, 2017. URL <http://arxiv.org/abs/1712.05690>.
- Edward J. Hu, Yelong Shen, Phillip Wallis, et al. LoRA: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021.
- Melvin Johnson, Mike Schuster, Quoc V. Le, Maxim Krikun, Yonghui Wu, Zhifeng Chen, Nikhil Thorat, Fernanda B. Viégas, Martin Wattenberg, Greg Corrado, Macduff Hughes, and Jeffrey Dean. “Google’s Multilingual Neural Machine Translation System: Enabling Zero-Shot Translation”. *arXiv*, 2016. URL <http://arxiv.org/abs/1611.04558>.

- Philipp Koehn and Rebecca Knowles. “Six Challenges for Neural Machine Translation”. *arXiv*, 2017. URL <http://arxiv.org/abs/1706.03872>.
- NLLB Team. No language left behind: Scaling human-centered machine translation. *arXiv preprint arXiv:2207.04672*, 2022.
- Ivan Provilkov, Dmitrii Emelianenko, and Elena Voita. BPE-Dropout: Simple and effective subword regularization. In *ACL*, 2020.
- Holger Schwenk, Vishrav Chaudhary, Shuo Sun, Hongyu Gong, and Francisco Guzmán. WikiMatrix: Mining 135m parallel sentences in 1620 language pairs from wikipedia. *EACL*, 2021.
- Rico Sennrich and Barry Haddow. Linguistic input features improve neural machine translation. In *WMT*, 2016.
- Rico Sennrich and Biao Zhang. “Revisiting Low-Resource Neural Machine Translation: A Case Study”. *arXiv*, 2019. URL <http://arxiv.org/abs/1905.11901>.
- Rico Sennrich, Barry Haddow, and Alexandra Birch. “Improving Neural Machine Translation Models with Monolingual Data”. *arXiv*, 2015. URL <http://arxiv.org/abs/1511.06709>.
- Rico Sennrich, Barry Haddow, and Alexandra Birch. “Neural Machine Translation of Rare Words with Subword Units”. *arXiv*, 2016. URL <http://arxiv.org/abs/1508.07909>.
- Nitish Srivastava, Geoffrey Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov. “Dropout: A Simple Way to Prevent Neural Networks from Overfitting”. *Journal of Machine Learning Research*, 15(56): 1929–1958, 2014. URL <http://jmlr.org/papers/v15/srivastava14a.html>.
- Brian Thompson and Philipp Koehn. Vecalign: Improved sentence alignment in linear time and space. In *EMNLP*, 2019.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. “Attention Is All You Need”. *arXiv*, 2017. URL <http://arxiv.org/abs/1706.03762>.
- Eryk Wdowiak. Sicilian translator: A recipe for low-resource machine translation. In *arXiv preprint arXiv:2110.01938*, 2021. doi: 10.1007/978-3-031-10464-0_50. URL <https://arxiv.org/abs/2110.01938>.